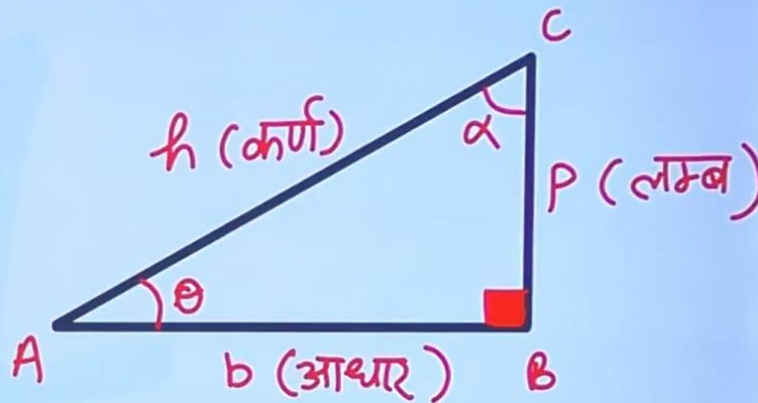
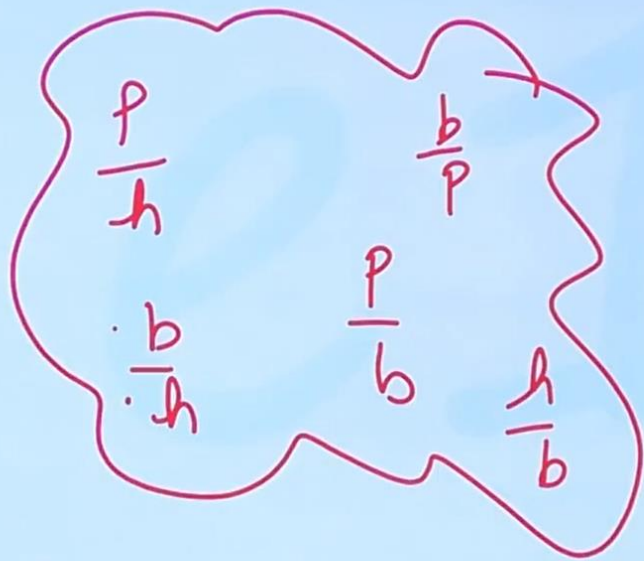


Trigonometric ratios (त्रिकोमितीय अनुपात):



Trigonometric ratios (त्रिकोणमितीय अनुपात):

Trigonometric functions:

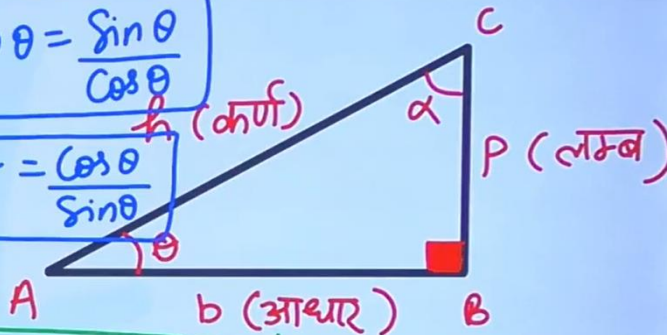
$$\sin \theta = \frac{P}{h}, \quad \operatorname{cosec} \theta = \frac{h}{P}$$

$$\cos \theta = \frac{b}{h}, \quad \sec \theta = \frac{h}{b}$$

$$\tan \theta = \frac{P}{b}, \quad \cot \theta = \frac{b}{P}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$



$$\sin \theta = \frac{1}{\operatorname{cosec} \theta}$$

$$\cos \theta = \frac{1}{\sec \theta}$$

$$\tan \theta = \frac{1}{\cot \theta}$$

$$\sin \theta \cdot \operatorname{cosec} \theta = 1$$

$$\cos \theta \cdot \sec \theta = 1$$

$$\tan \theta \cdot \cot \theta = 1$$

Write the answers of the following (Forward and backward):

$\sin \theta$	$\frac{b}{h}$	$\operatorname{cosec} \theta$	$\sec \theta$
$\frac{p}{b}$	$\cot \theta$	$\frac{h}{b}$	$\cos \theta$
$\sin \theta$	$\frac{h}{p}$	$\tan \theta$	$\frac{p}{h}$
$\frac{b}{p}$	$\cos \theta$	$\operatorname{cosec} \theta$	$\sec \theta$
$\tan \theta$	$\cot \theta$	$\frac{p}{h}$	$\frac{b}{p}$
$\frac{h}{p}$	$\frac{p}{b}$	$\frac{h}{b}$	$\frac{b}{h}$

If $\sin\theta = \frac{3}{5}$, then $\frac{\tan\theta + \cos\theta}{\cot\theta + \operatorname{cosec}\theta} = ?$

अगर $\sin\theta = \frac{3}{5}$ है, तो $\frac{\tan\theta + \cos\theta}{\cot\theta + \operatorname{cosec}\theta} = ?$

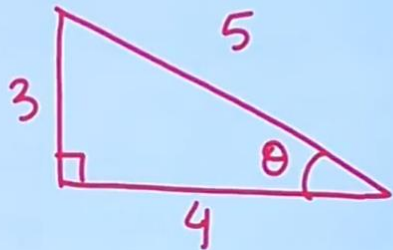
a) $\frac{29}{60}$

b) $\frac{31}{60}$

c) $\frac{34}{60}$

d) $\frac{37}{60}$

✓
 $\sin\theta = \frac{p}{h} = \frac{3}{5}$



$$\frac{\frac{3}{4} + \frac{4}{5}}{\frac{4}{3} + \frac{5}{3}} = \frac{31}{20 \times 3} = \frac{31}{60}$$

If $5\tan\theta = 4$, then $\frac{5\sin\theta - 3\cos\theta}{5\sin\theta + 2\cos\theta}$ is

अगर $5\tan\theta = 4$ है, तो $\frac{(5\sin\theta - 3\cos\theta)/\cos\theta}{(5\sin\theta + 2\cos\theta)/\cos\theta}$ है?

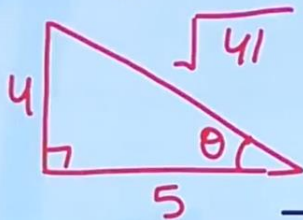
a) $\frac{2}{3}$

b) $\frac{1}{4}$

☒ c) $\frac{1}{6}$

d) $\frac{1}{3}$

$\tan\theta = \frac{4}{5} = \frac{p}{b}$



$5 \times \frac{4}{\sqrt{41}}$

$\frac{5 \times \frac{p}{h} - 3 \times \frac{b}{h}}{\quad}$

$\frac{5 \times 4 - 3 \times 5}{5 \times 4 + 2 \times 5}$

$\frac{5 \times \frac{p}{h} + 2 \times \frac{b}{h}}{\quad} = \frac{5}{30} = \frac{1}{6}$

If $5\tan\theta = 4$, then $\frac{5\sin\theta - 3\cos\theta}{5\sin\theta + 2\cos\theta}$ is

अगर $5\tan\theta = 4$ है, तो $\frac{5\sin\theta - 3\cos\theta}{5\sin\theta + 2\cos\theta} = ?$

a) $\frac{2}{3}$

b) $\frac{1}{4}$

c) $\frac{1}{6}$

d) $\frac{1}{3}$

$$\frac{\sin\theta}{\cos\theta} = \frac{4}{5}$$

$$\underline{5\sin\theta = 4\cos\theta}$$

$$\frac{\cancel{1\cos\theta}}{\cancel{6\cos\theta}} = \frac{1}{6}$$

If $\sin \theta = \frac{a}{b}$, then the value of $\sec \theta - \cos \theta$ is (where $0^\circ < \theta < 90^\circ$)

अगर $\sin \theta = \frac{a}{b}$ है, तो $\sec \theta - \cos \theta$ होगा- (जब $0^\circ < \theta < 90^\circ$)

a) $\frac{a}{b\sqrt{b^2-a^2}}$

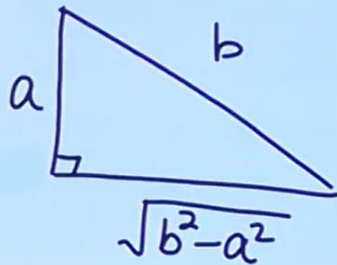
b) $\frac{b^2}{a\sqrt{b^2-a^2}} \cdot \frac{1}{b}$

c) $\frac{a^2}{b\sqrt{b^2-a^2}}$

d) $\frac{\sqrt{b^2+a^2}}{\sqrt{b^2-a^2}}$

$$\sin \theta = \frac{p}{h} = \frac{a}{b}$$

$$\begin{aligned} & \frac{b}{\sqrt{b^2-a^2}} - \frac{\sqrt{b^2-a^2}}{b} \\ &= \frac{a^2}{b\sqrt{b^2-a^2}} \end{aligned}$$



If $\frac{\sin\theta}{x} = \frac{\cos\theta}{y}$, then $\sin\theta - \cos\theta$ is equal to

अगर $\frac{\sin\theta}{x} = \frac{\cos\theta}{y}$ है, तो $\sin\theta - \cos\theta = ?$

a) $x - y$

b) $x + y$

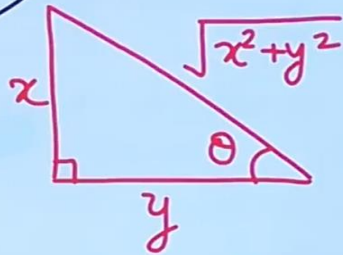
c) $\frac{x-y}{\sqrt{x^2+y^2}}$

d) $\frac{(y-x)}{\sqrt{x^2+y^2}}$

$$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{x}{y} = \frac{p}{b}$$

$$\frac{p}{h} - \frac{b}{h} = \frac{p-b}{h}$$

$$= \frac{x-y}{\sqrt{x^2+y^2}}$$



If $5\sin\theta - 4\cos\theta = 0, 0^\circ < \theta < 90^\circ$, then the value of $\frac{5\sin\theta - 2\cos\theta}{5\sin\theta + 3\cos\theta}$ is:

यदि $5\sin\theta - 4\cos\theta = 0, 0^\circ < \theta < 90^\circ$ है, तो $\frac{(5\sin\theta - 2\cos\theta)/\cos\theta}{(5\sin\theta + 3\cos\theta)/\cos\theta}$ का मान है:

a) $\frac{3}{8}$

b) $\frac{3}{7}$

c) $\frac{2}{7}$

d) $\frac{5}{8}$

9878125144@gmail.com 9878125144

① $5\sin\theta = 4\cos\theta$

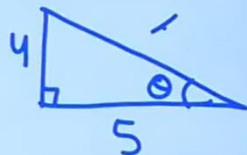
$$\frac{2\cancel{\cos\theta}}{7\cancel{\cos\theta}} = \frac{2}{7}$$

② $\tan\theta = \frac{4}{5}$

$$= \frac{5\tan\theta - 2}{5\tan\theta + 3}$$

$$= \frac{2}{7} \checkmark$$

③ $\tan\theta = \frac{4}{5}$



$$\frac{5 \times 4 - 2 \times 5}{5 \times 4 + 3 \times 5} = \frac{2}{7}$$

If $2y \cos \theta = x \sin \theta$ and $2x \sec \theta - y \operatorname{cosec} \theta = 3$, then the value of $x^2 + 4y^2$ is

यदि $2y \cos \theta = x \sin \theta$ और $2x \sec \theta - y \operatorname{cosec} \theta = 3$ हो, तो

$x^2 + 4y^2$ का मान क्या होगा?

a) 1

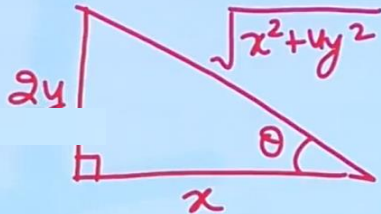
b) 2

c) 3

d) 4

$h^2 = 4$

$$\tan \theta = \frac{2y}{x} = \frac{p}{b}$$



$$2x \times \frac{h}{x} - y \times \frac{h}{2y} = 3$$

$$= \frac{2}{2} h = 3$$
$$h = 2$$

If $0 < \theta < 90^\circ$, $3b \operatorname{cosec} \theta = a \sec \theta$, and $3a \sec \theta - b \operatorname{cosec} \theta = 8$, then the value of $9b^2 + a^2$ is:

यदि $0 < \theta < 90^\circ$, $3b \operatorname{cosec} \theta = a \sec \theta$ और $3a \sec \theta - b \operatorname{cosec} \theta = 8$ है, तो $9b^2 + a^2$ का मान ज्ञात करें।

a) 6

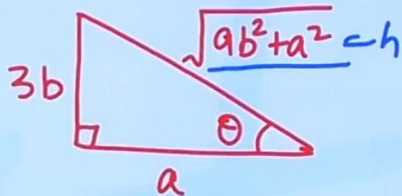
$h^2 = 9$

b) 8

c) 9

d) 7

$$\tan \theta = \frac{3b}{a} = \frac{\sin \theta}{\cos \theta}$$



$$3\cancel{a} \times \frac{h}{\cancel{a}} - \cancel{b} \times \frac{h}{\cancel{3b}} = 8$$

$$\Rightarrow \frac{\cancel{3}h}{\cancel{3}} = \cancel{8}$$

$$\textcircled{h=3}$$

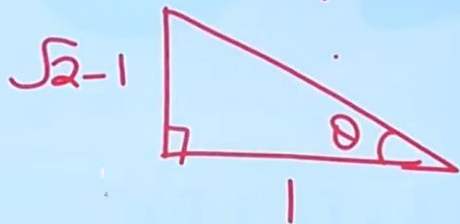
If $\sec \theta (\cos \theta + \sin \theta) = \sqrt{2}$, then what is the value of $\frac{2 \sin \theta}{\cos \theta - \sin \theta}$?

यदि $\sec \theta (\cos \theta + \sin \theta) = \sqrt{2}$ है तो $\frac{2 \sin \theta}{\cos \theta - \sin \theta}$ का मान क्या है?

- a) $3\sqrt{2}$ b) $\frac{3}{\sqrt{2}}$ c) $\frac{1}{\sqrt{2}}$ d) $\sqrt{2}$

$$1 + \tan \theta = \sqrt{2}$$

$$\tan \theta = \frac{\sqrt{2} - 1}{1} = \frac{p}{b}$$



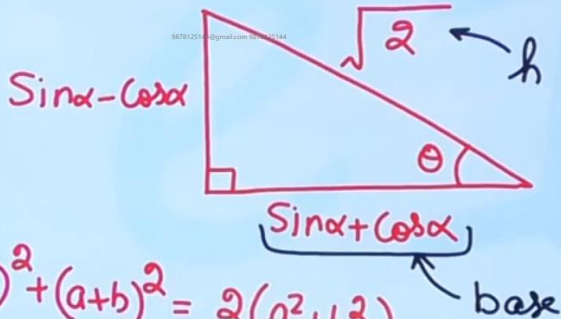
$$\frac{2 \times \frac{p}{h}}{\frac{b}{h} - \frac{p}{h}}$$

$$\textcircled{\sqrt{2}} = \frac{2 \times \sqrt{2} - 1}{\cancel{(\sqrt{2}-1)} + \cancel{\sqrt{2}}(\sqrt{2}-1)}$$

If $\tan\theta = \frac{\sin\alpha - \cos\alpha}{\sin\alpha + \cos\alpha}$ then $(\sin\alpha + \cos\alpha)$ is

अगर $\tan\theta = \frac{\sin\alpha - \cos\alpha}{\sin\alpha + \cos\alpha}$ है, तो $(\sin\alpha + \cos\alpha) = ?$

- a) $\pm\sqrt{2}\sin\theta$ ~~b) $\pm\sqrt{2}\cos\theta$~~ c) $\pm\frac{1}{\sqrt{2}}\sin\theta$ d) $\pm\frac{1}{\sqrt{2}}\cos\theta$



$$\begin{aligned} (a-b)^2 + (a+b)^2 &= 2(a^2 + b^2) \\ / \\ &= 2(\sin^2\alpha + \cos^2\alpha) \\ &= 2 \end{aligned}$$

$$\cos\theta = \frac{\sin\alpha + \cos\alpha}{\sqrt{2}}$$

$$\sqrt{2} \cos\theta = \sin\alpha + \cos\alpha$$

In a $\triangle ABC$, right angled at B , $AB = 7$ cm and $(AC - BC) = 1$ cm, the value of $(\sec C + \cot A)$ is:

$\triangle ABC$ में, B पर समकोण है, $AB = 7$ cm और $(AC - BC) = 1$ cm है। $(\sec C + \cot A)$ का मान है:

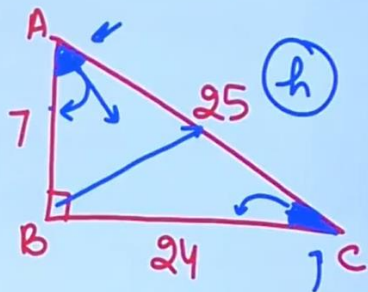
a) 1

b) $\frac{19}{24}$

c) $\frac{3}{4}$

d) $\frac{4}{3}$

$$\frac{4}{3} = \frac{32}{24} = \frac{25}{24} + \frac{7}{24}$$



If $\tan^2 \theta = 1 - e^2$ then the value of $\sec \theta + \tan^3 \theta \operatorname{cosec} \theta$ is

अगर $\tan^2 \theta = 1 - e^2$ है, तो $\sec \theta + \tan^3 \theta \operatorname{cosec} \theta = ?$

- a) $(2 + e^2)^{\frac{3}{4}}$ b) $(2 - e^2)^{\frac{1}{2}}$ c) $(2 + e^2)^{\frac{1}{2}}$ d) $(2 - e^2)^{\frac{3}{2}}$

$$(\sqrt{3})^3 = 3\sqrt{3}$$

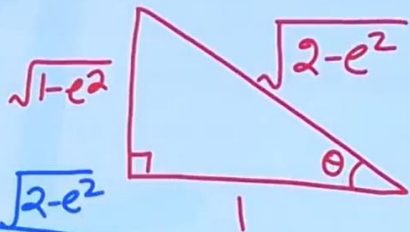
$$\tan \theta = \frac{\sqrt{1-e^2}}{1}$$

$$\frac{\sqrt{2-e^2}}{1} + \frac{(1-e^2) \cancel{\sqrt{1-e^2}} \times \sqrt{2-e^2}}{\cancel{\sqrt{1-e^2}}}$$

$$= \sqrt{2-e^2} (1 + 1 - e^2)$$

$$= \sqrt{2-e^2} \times (2-e^2)^1$$

$$= (2-e^2)^{3/2}$$



$$\sqrt{1-e^2+1}$$

If $\tan^2 \theta = 1 - e^2$ then the value of $\sec \theta + \tan^3 \theta \operatorname{cosec} \theta$ is

अगर $\tan^2 \theta = 1 - e^2$ है, तो $\sec \theta + \tan^3 \theta \operatorname{cosec} \theta = ?$

- a) $(2 + e^2)^{\frac{3}{4}}$ b) $(2 - e^2)^{\frac{1}{2}}$ c) $(2 + e^2)^{\frac{1}{2}}$ d) $(2 - e^2)^{\frac{3}{2}}$

$$\sec^2 \theta = (2 - e^2)$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$\sec \theta = \sqrt{2 - e^2}$$

$$\sec \theta + \frac{\sin^2 \theta}{\cos^2 \theta} \times \sec \theta$$

① दिया है

② सिद्धा है

$$= (2 - e^2)^{3/2} = \sec^3 \theta \times \sec^2 \theta$$

$$= \sec^3 \theta$$

$$=$$

In a $\triangle ABC$, $\angle C = 90^\circ$ then the equation, whose roots are $\tan A$ and $\tan B$, is

$\triangle ABC$ में, $\angle C = 90^\circ$ है, तो वह समीकरण बताइए जिसके मूल $\tan A$ और $\tan B$ हैं।

a) $abx^2 + c^2x + ab = 0$

c) $abx^2 - c^2x - ab = 0$

b) $abx^2 + c^2x - ab = 0$

d) $abx^2 - c^2x + ab = 0$

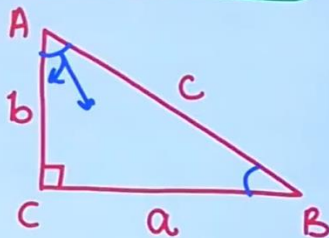
$x^2 - (S \text{ or } R)x + P \text{ or } R = 0$

$A + \tan B$

$\frac{a}{b} + \frac{b}{a} = \frac{a^2 + b^2}{ab} = \frac{c^2}{ab}$

$\tan A \times \tan B$

$= \frac{a}{b} \times \frac{b}{a} = 1$



$x^2 - \frac{c^2}{ab}x + 1 = 0$

$\Rightarrow abx^2 - c^2x + ab = 0$

Assume the earth to be a sphere of radius R . What is the radius of the circle of latitude 40°S ?

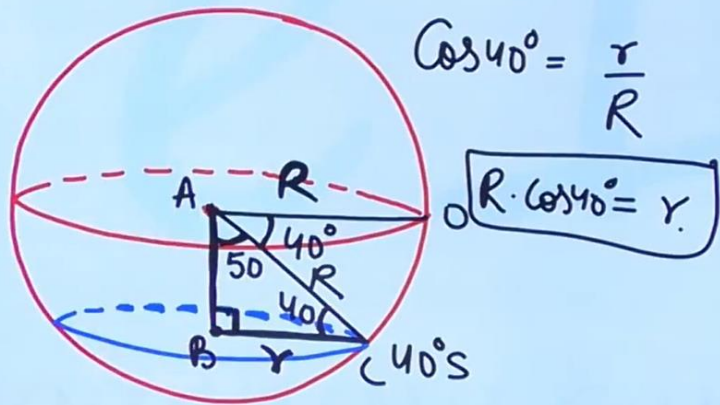
पृथ्वी को त्रिज्या R वाला एक गोला मान लें। अक्षांश 40°S के वृत्त की त्रिज्या क्या है?

☒ a) $R \cos 40^\circ$

b) $R \sin 80^\circ$

c) $R \sin 40^\circ$

d) $R \tan 40^\circ$



$\left\{ \begin{array}{l} \sin 20^\circ = \cos 70^\circ \\ \sin 30^\circ = \cos 60^\circ \\ \sin 10^\circ = \cos 80^\circ \end{array} \right.$